

Electronic Card Lock for Appliances

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The circuit presented here can be used as a lock for important electronic and electrical appliances. When a card is inserted inside

its mechanism, depending upon the positions of the holes punched on the card, a particular appliance is switched on. The card is inserted just like a CD is pushed inside the disk drive. It is rectangular in shape with holes punched on it.

The electronic lock is connected to an access control system, which ensures that only the selected appliance is switched on

and prevents false operation of the system.

Circuit description

Fig. 1 shows the block diagram of the electronic card lock. The lock has three main parts: Card sensor, microcontroller and appliance (load). The card sensor senses the code on the card (holes punched on the card) and provides the signal to the microcontroller. The microcontroller compares the code and switches on the corresponding device.

Fig. 2 shows the circuit of the electronic card lock. It comprises

quad optocoupler ILQ74, microcontroller AT89C51, regulator 7805 and a few discrete components. The card sensor consists of eight light-dependent resistors (LDRs), eight LEDs and two switches (S2 and S3). LDR1 through LDR8 and LED1 through LED8 are placed face to face, respectively. One terminal of all the LDRs is connected to 5V and the other terminal is connected to the respective LED of

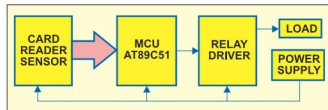


Fig. 1: Block diagram of electronic card lock using microcontroller

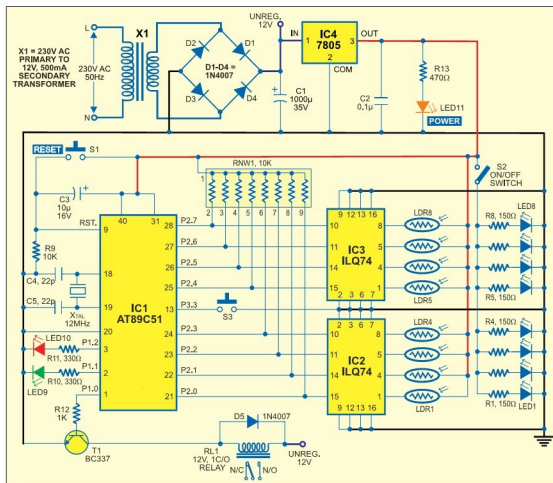


Fig. 2: Circuit of electronic card lock using microcontroller



Fig. 3: Pin configurations of 7805 and BC337